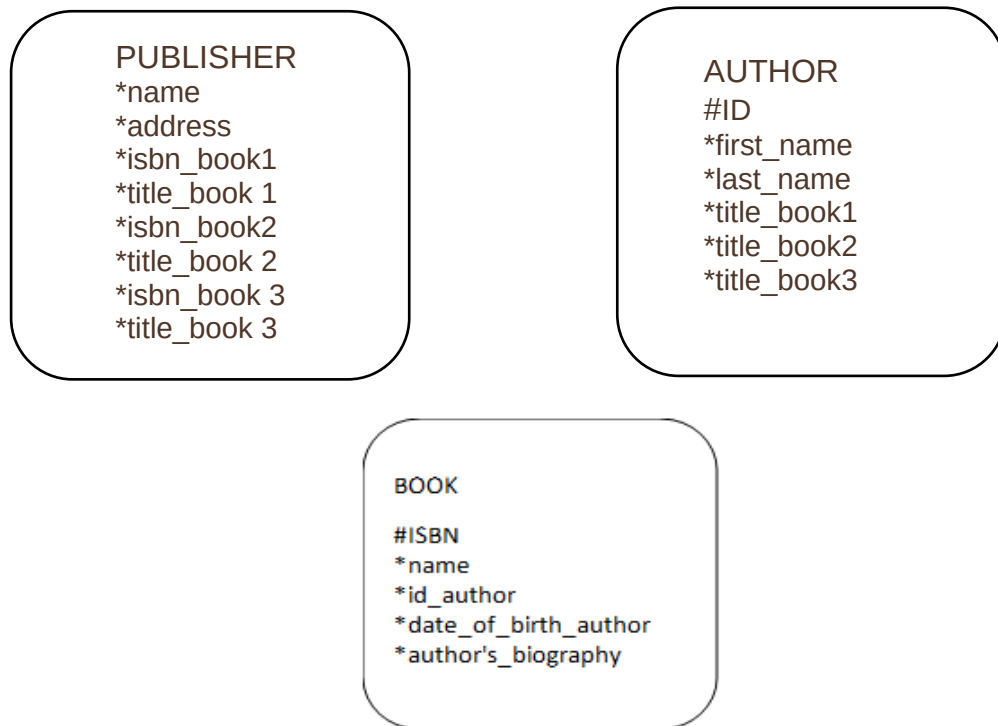


TASKS EXDON 09.12.2024
Exam preparation

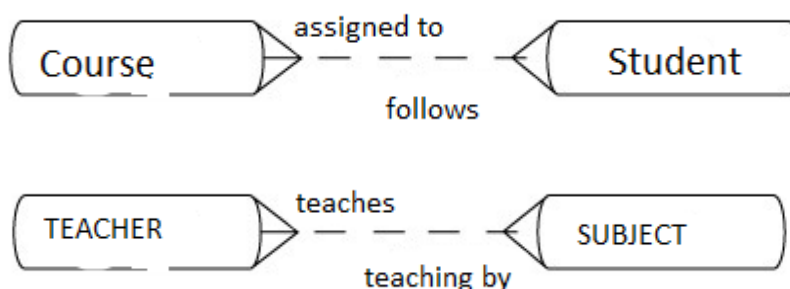
Exercise A: Draw ERDs for each of the following. Draw softboxes, relationship lines, and labels for each relationship in both directions. Indicate non-transferability when appropriate.

- a) Each EMPLOYEE may be assigned to one and only one PROJECT. Each PROJECT must have assigned one or more employees.
- b) Each PRODUCT must have one and only one PRICE. Each PRICE may be for one or more products.
- c) Each SCHOOL may be attended by one or more STUDENTS. Each STUDENT must attend one and only one SCHOOL.
- d) Each EMPLOYEE must hold one and only one JOB. Each JOB may be held by one or more EMPLOYEEs.
- e) Each TOWN may be the birthplace of many PEOPLE. Each PERSON must be born in one and only one TOWN.

Exercise B: Are the following entities normalized? If not, make the necessary changes to correct it.



Exercise C: Resolve the M:M relationships.



Exercise D:

1. Determine how exclusive relationships should be modeled in the following scenario.

An Event can be held at either a Private Home or a Public Space.

2. Develop a hierarchical structure to represent the following situation.

Each President can have one or more Directors. Each Director can have one or more Managers.

Each Manager can have one or more Employees.

Exercise E: Using the following scenario

- 1. Draw the conceptual model**
- 2. Identify the possible Entities, their Attributes and the relationships between the entities from the given scenario.**
- 3. Draw the ERD following diagramming convention.**
- 4. Draw the physical model**
- 5. Write the sentences for each relationship, in both directions.**
- 6. Map the relationships.**
- 7. Draw the matrix diagram.**

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